

Synopsis of Cohort and Decile Analysis of Customers

The purpose of this analysis is to study customer purchasing behavior and performance across different groups using cohort and decile analysis. The primary goal is to understand how customers differ in purchase frequency, purchasing behavior, and profitability. By analyzing metrics such as average order frequency (AOF), average order value (AOV), and average margin, the analysis identifies which customer segments generate the most profit and how customer performance changes over time across different cohorts.

Decile analysis divides customers into ten equal groups based on their value. Each decile includes approximately 3,180 customers, allowing for comparison of purchasing behavior across different segments. The results show a strong concentration of value among the most valuable customers. The first decile, representing the most valuable customers, has an average order frequency of 5.50, significantly higher than all other groups. These customers also generate the highest spend, with an average order value of 7,367.45 and an average margin of 48.2%.

In comparison, customer activity in the lower deciles declines. For example, in the second decile, the average activity factor (AOF) is 2.74 and the average order value (AOV) is 6,382.04, while in the third decile, the AOF is 1.99 and the AOV is 6,032.19. By the fifth decile, the average order value (AOV) drops to 4,816.41 and the AOF ratio to 1.47. The segment with the lowest value, the tenth decile, exhibits much lower purchase intensity: the AOF is only 1.03 and the AOV is 1,143.90, with the average margin also falling to 41.6%. These results indicate that a relatively small portion of customers generates significantly higher purchase activity and revenue compared to the rest of the customer base.

Cohort analysis examines groups of customers based on the quarter in which they were acquired to observe trends in behavior over time. The largest cohort in the dataset is 2021Q1 with 2,943 customers, followed by smaller acquisition cohorts such as 2021Q2 with 346 customers and 2021Q3 with 313 customers. Across all cohorts, the average order value remains relatively high but varies over time. For example, customers acquired in 2021Q1 show an AOV of 5920.17 with an AOF of 1.25, while the 2021Q4 cohorts show stronger engagement with an AOF of 1.69 and an AOV of 5730.07. Later cohorts show moderate fluctuations, such as 2022Q4, which achieved one of the highest order frequencies with an AOF of 1.92 and an AOV of 5716.80.

Overall, the analysis reveals two important patterns. First, customer value is distributed extremely unevenly: customers in the top decile purchase much more frequently and spend significantly more per order. Second, cohort metrics show that newer customer groups can sometimes be more engaged than older ones, suggesting that customer acquisition strategies or customer behavior may improve over time. These findings can help companies focus their customer retention strategies on high-value segments while improving engagement with low-value customers to increase overall revenue and profitability.

